Learning Objectives 9.2–9.4: I can articulate what a Uniform Plane Wave (UPW) is and describe key properties that relate the electric and magnetic field. I can calculate the electric and magnetic fields for a UPW. I can articulate the concept of polarization and determine the polarization for various UPWs.

Show your work. Use the IEEE convention for handedness. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **A plane wave in a dielectric.** A uniform plane wave travels along $+z$ in a lossless dielectric with $\epsilon_r = 4$ and $\mu = \mu_0$. Its electric field is

$$\mathbf{E} = \hat{\mathbf{x}}\, 6 \cos\left(2\pi \times 10^8\, t - \beta z\right) \text{ V/m}$$

   (a) Starting from Faraday's law in phasor form, $\nabla \times \widetilde{\mathbf{E}} = -j\omega\mu\widetilde{\mathbf{H}}$, derive $\widetilde{\mathbf{H}}$ for $\widetilde{\mathbf{E}} = \hat{\mathbf{x}}\, E_0 e^{-j\beta z}$, and show that $E_0/H_0 = \sqrt{\mu/\epsilon}$.

   (b) Find β, λ, v_p, and η.

β =

λ =

v_p =

η =

   (c) Write $\mathbf{H}(z,t)$.

2. **A wave along y.** In free space,

$$\mathbf{E} = \hat{\mathbf{z}}\, 3 \cos\left(2\pi \times 10^9\, t - \beta y\right) \text{ V/m}$$

(a) Give the direction of travel and find β.

$\hat{\mathbf{k}}$ =

β =

(b) Find $\mathbf{H}(y, t)$, and check that $\mathbf{E} \times \mathbf{H}$ points along the direction of travel.

3. **Identify the polarization.** For each wave, state whether it is linear, circular, or elliptical, give the handedness of a circular or elliptical wave, and give the tilt from the x axis of a linear one.

   (a) $\widetilde{\mathbf{E}} = (\hat{\mathbf{x}}\,3 + \hat{\mathbf{y}}\,4)\,e^{-j\beta z}$

   (b) $\widetilde{\mathbf{E}} = (\hat{\mathbf{x}} - \hat{\mathbf{y}})\,e^{-j\beta z}$

   (c) $\widetilde{\mathbf{E}} = 5\,(\hat{\mathbf{x}} + j\hat{\mathbf{y}})\,e^{-j\beta z}$

   (d) $\mathbf{E} = \hat{\mathbf{x}}\cos(\omega t - \beta z) + \hat{\mathbf{y}}\,2\sin(\omega t - \beta z)$

   (e) $\widetilde{\mathbf{E}} = (j\hat{\mathbf{x}} + \hat{\mathbf{y}})\,e^{+j\beta z}$

4. **The axes of an ellipse.** A wave along $+z$ has, at $z = 0$,

$$\mathbf{E} = \hat{\mathbf{x}} \cos \omega t + \hat{\mathbf{y}} \cos\left(\omega t + 60^\circ\right) \text{ V/m}$$

(a) Starting from $E_x^2 - 2E_x E_y \cos\delta + E_y^2 = \sin^2\delta$, find $|\mathbf{E}|$ where the path crosses the lines $y = x$ and $y = -x$.

(b) Give the tilt of the major axis, the axial ratio, and the handedness.

tilt = ____

axial ratio = ____

hand = ____

Documentation: _____